



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
PART TEST
29-11-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS, TOAS, TAAS & TNAS] (SCORE-I)

Time : 3 Hours

PAPER-1

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS
SECTION-I (i) : (Maximum Marks: 12)

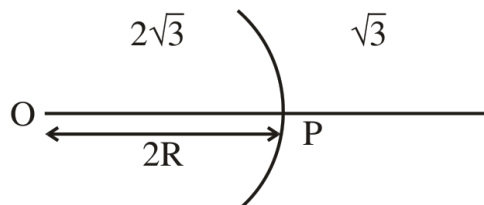
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

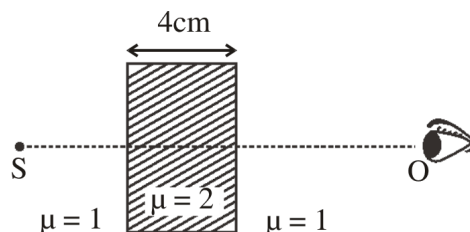
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. A point source O kept at distance of $2R$ from pole P of concave interface having radius of curvature R emits a unidirectional beam at an angle 30° to the principal axis PO. If the distance of point of intersection of refracted ray principal axis after refraction from pole P is:



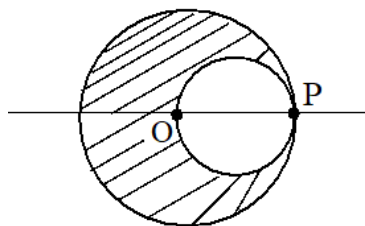
- (A) R (B) $2R$ (C) $3R$ (D) $4R$
2. An object S is placed at a distance d from the left face of the slab having refractive index $= 2$. The image is observed by an observer O. The image is



- (A) The image is real and shift towards observer by 2cm.
- (B) The image is virtual and shift away from observer by 2cm.
- (C) The image is virtual and shift towards observer by 2cm.
- (D) The image is real and shift away from observer by 2cm.

Space for Rough Work

3. A transparent sphere of radius R has a cavity of radius $R/2$ as shown in figure. Find the refractive index of the sphere if a parallel beam of light falling on left surface gets focused at point P .



- (A) $\mu = \frac{3 + \sqrt{5}}{2}$ (B) $\mu = \frac{3 - \sqrt{5}}{4}$ (C) $\mu = 3 + \sqrt{5}$ (D) $\mu = \frac{1 + \sqrt{5}}{2}$
4. A solid sphere toy globe of mass M and radius R rotates freely without friction with an initial angular velocity ω_0 . A bug of mass m starts at the north pole and travels with constant relative speed v to the south pole along a meridian. If the axis of rotation of the globe is held fixed then angular velocity of the globe as a function of time is :

- (A) $\omega = \frac{2M\omega_0}{2M + 5m\sin^2(vt/R)}$ (B) $\omega = \frac{2M\omega_0}{2M + 5m\cos^2(vt/R)}$
- (C) $\omega = \frac{5M\omega_0}{2M + 5m\cos^2(vt/R)}$ (D) $\omega = \frac{5M\omega_0}{2M + 5m\sin^2(vt/R)}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

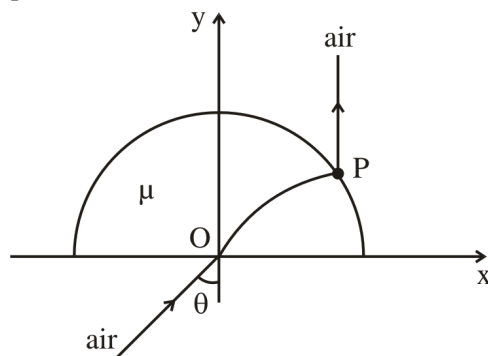
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

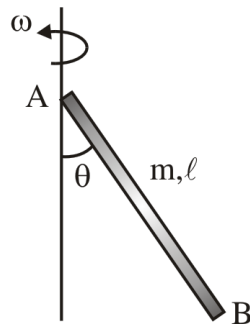
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. A hemisphere of radius $a/2$ and made up of a material of variable refractive index is placed with its base centre O at the origin as shown in the figure. The refractive index of the material of the hemisphere varies as $\mu = \left(\frac{a}{a-x} \right)$. A ray of light is incident at the point O at an angle $\theta (= 0)$ with the normal in the x-y plane and it comes out through a point P on its curved surface. Then choose the correct option(s).



- (A) The equation of trajectory of ray of light inside the hemisphere is $y = \sqrt{2ax - x^2}$.
- (B) The equation of trajectory of ray of light inside the hemisphere is $y = \sqrt{ax - 4x^2}$.
- (C) The x and y coordination of point P are $\left(\frac{a}{8}, \frac{a}{8}\sqrt{15} \right)$.
- (D) The x and y coordination of point P are $\left(\frac{a}{2}, \frac{a}{2}\sqrt{15} \right)$.

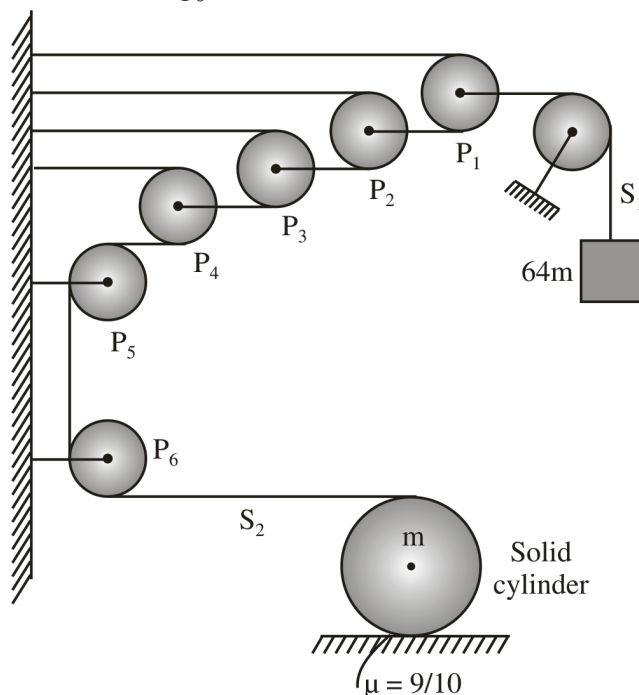
6. A thin uniform rod AB of mass $m = 1\text{ kg}$ and length $\ell = 1\text{ m}$ is hinged at end A. The rod makes an angle $\theta = 53^\circ$ with the vertical axis and it is rotating with a constant angular velocity ω about the vertical axis passing through end A as shown in the figure. Then choose the correct option(s).



- (A) The angular velocity ω must be 5 rad/s .
 (B) The angular velocity ω must be 10 rad/s .
 (C) The net hinge reaction on the rod at end A $10\sqrt{2}\text{ N}$.
 (D) The net hinge reaction on the rod at end A 10 N .

Space for Rough Work

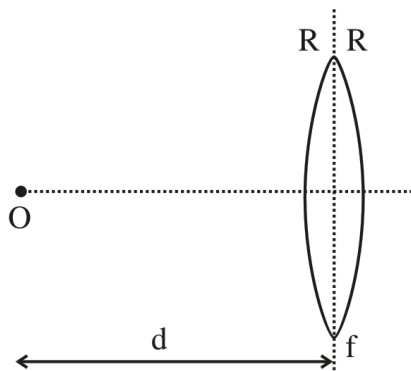
7. All the pulleys and strings are massless. One end of the string S_1 is connected with a block and one end of the string S_2 is connected with a solid cylinder as shown in the figure. The coefficient of friction between the cylinder and the horizontal surface is $\mu = \frac{9}{10}$. Choose the correct option(s). (Taken $m = 2\text{kg}$ and $g = 10\text{ m/s}^2$)



- (A) The tension in the string attached with the block $64m$ is 48 N .
 (B) The acceleration of block $64m$ is 4 m/s^2 downward.
 (C) The acceleration of centre of mass of the solid cylinder is 32 m/s^2 .
 (D) Friction force between the cylinder and the horizontal surface is 16 N .

Space for Rough Work

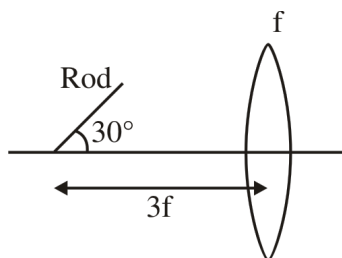
8. Two insects each of mass m are at opposite ends of a diameter of a turntable of radius R . Moment of inertia of the turntable about its central vertical fixed axis is $I = \frac{2}{3}mR^2$. The turntable is rotating freely with angular speed ω and both the insects start crawling slowly towards each other on the turntable. Then.
- (A) The insect have to apply maximum horizontal force on the turntable when they are at a distance $R/2$ from the centre of the turntable.
- (B) The insect have to apply maximum horizontal force on the turntable when they are at a distance $R/3$ from the centre of the turntable.
- (C) The angular speed of the turntable when the insects exert maximum force on the turntable equal 3ω .
- (D) The maximum force exerted by an insect on the turntable during the motion is equal to $3m\omega^2R$.
9. A thin equiconvex lens made of a material having refractive index $\mu_L = \frac{3}{2}$, has a focal length f in air. Image of an object placed in front of it, is inverted, real and magnified. Now the whole arrangement is immersed in water ($\mu_W = \frac{4}{3}$) without changing distance between the object and the lens. Then



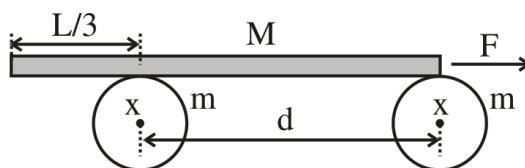
- (A) The focal length of lens relative to water will be $4f$.
- (B) New image will be virtual, erect and magnified.
- (C) New image will be real, inverted and diminished.
- (D) If the lens is cut laterally and then longitudinally into four identical parts (inside water) then the focal length of each part will be $8f$.

Space for Rough Work

10. A thin convex lens of focal length f is having small rod at its optical axis at $3f$ distance in front of it. [angular velocity given is about axis perpendicular to plane of paper]



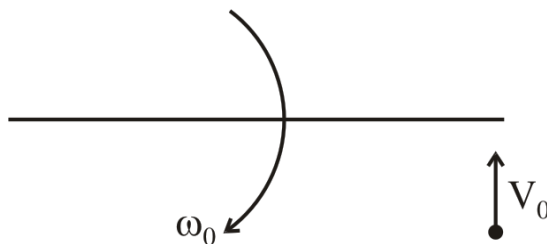
- (A) Image of rod is inclined at $\tan^{-1} \left(\frac{2}{\sqrt{3}} \right)$ angle with optical axis.
- (B) If rod is rotated with angular velocity $\omega = 7$ rad/sec its image will start to rotate with angular velocity 8 rad/s
- (C) If lens is rotated with angular velocity 2 rad/sec then image of rod will rotate with angular velocity 15 rad/s
- (D) length of image of rod is smaller than length of the rod.
11. A rigid rod of length L and mass M , whose mass is distributed uniformly, is placed on two identical thin-walled cylinders resting on a rod, one of its endpoints is directly above the axis of one cylinder, while its trisector point (closer to its other end) is directly above the axis of the other cylinder. The mass of the cylinder is m each. A constant horizontal pulling force F acts on the rod. Both cylinders roll without friction.



- (A) The speed of Rod when its leftmost end is exactly above the axis of left cylinder is $\sqrt{\frac{8FL}{3(M+m)}}$
- (B) The speed of Rod when its leftmost end is exactly above the axis of left cylinder is $\sqrt{\frac{4FL}{3(M+m)}}$
- (C) The minimum coefficient of friction required between cylinder & Rod for pure rolling is $\frac{2mF}{3(M+m)Mg}$
- (D) The minimum coefficient of friction required between cylinder and Rod for pure rolling is $\frac{2mF}{(M+m)Mg}$

Space for Rough Work

12. A uniform rod of mass $m_r = 2$ kg and length $l = 1$ m is spinning on a frictionless table with an angular speed $\omega_0 = 6$ rad/s (the centre of the rod is not translating but it is free to move). A particle of mass $m_p = 1$ kg moving with a velocity $v_0 = 6$ m/s, collides at the end of the rod in the perpendicular direction (see figure). If coefficient of restitution of the collision is $e = 2/3$ then which of the following is correct?



- (A) The particle keeps on moving in the same direction with a speed 1 m/s.
 (B) The rod start rotating in the opposite direction with an angular speed 9 rad/s.
 (C) The centre of mass of the rod starts moving with a speed 2.5 m/s.
 (D) The rod continues to rotate in the same direction but with a reduced angular speed.

Space for Rough Work

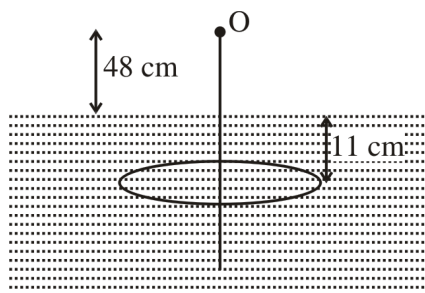
SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

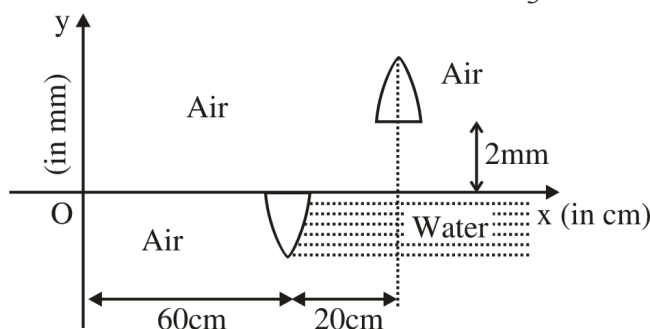
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. A convex lens has a focal length 15 cm in air. The refractive index of lens is 1.5. The lens is placed inside the water ($\mu = 4/3$) at a depth 11 cm below the water surface. A point object is placed in air on the principal axis of the lens at a height 48 cm above the water surface as shown in figure. The distance (in m) of final image of the object from the water surface, is :

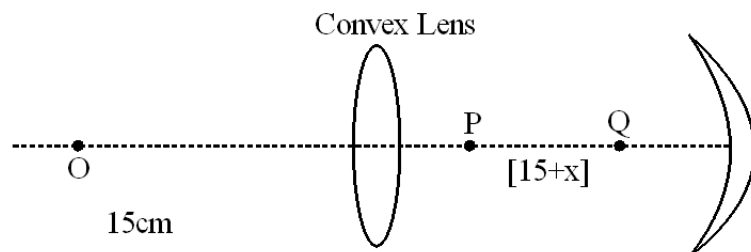


2. A point object 'O' is placed at the origin a thin equi-convex glass lens (refractive index of glass is 1.5) and its focal length in air medium is 20cm and placed so that its principal axis along x-axis. Now the lens is cut at the middle (along the principal axis) and upper half is shifted along x-axis and y-axis by 20cm and 2mm respectively and right side of lower half is filled with water ($\mu_w = 4/3$) as shown in figure then the difference in y-coordinate of the images formed by the upper half and lower half is $\frac{x}{3}$ mm, then the value of 'x' is

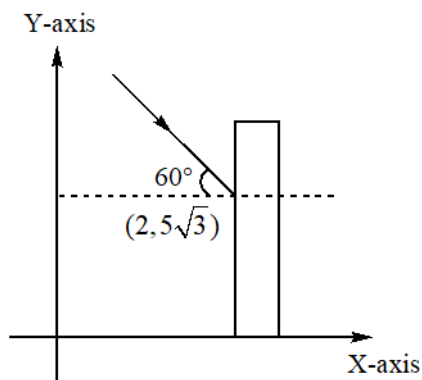


Space for Rough Work

3. A bright point object is placed at a distance of 15 cm in front of a fixed convex lens of focal length 10 cm (on its principal axis). A concavo convex lens of refractive index 1.5 with radii of curvature 60cm and 30cm can move coaxially behind the convex lens. The convex surface of the concavo - convex lens is silvered. It is observed that at two positions P and Q of the silvered lens, the final image coincides with the object. If the distance between P and Q is $[15+x]$ cm, find the value of x.

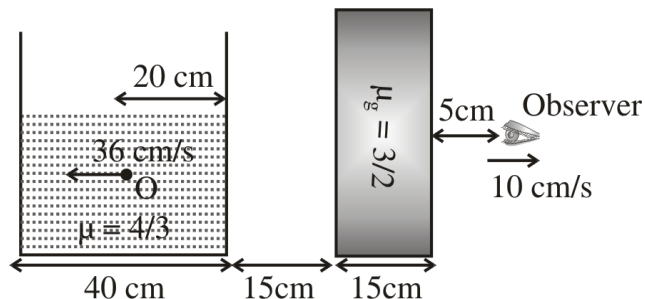


4. A slab of glass of thickness of 3 unit and refractive index $\sqrt{3}$ is surrounded by vacuum and is placed as shown. A light ray is incident at the point $(2, 5\sqrt{3})$ as shown. The point where the ray will cut on the x-axis is $(x, 0)$. Find the value of x:

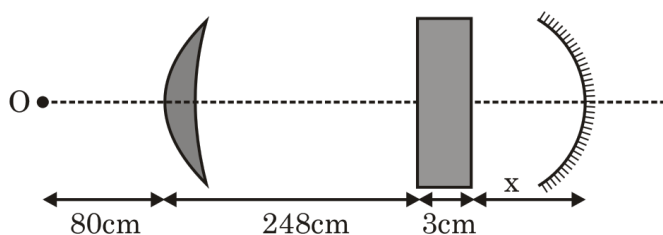


Space for Rough Work

5. A point object O lies inside a transparent cuboidal container (of negligible thickness) having water as shown in the figure. An observer is situated between the glass slab. At $t = 0$ observer starts moving with a velocity 10 cm/s towards right and object O starts moving towards left side with a velocity 36 cm/s as shown in the figure. Then the velocity (in cm/sec) of image of object with respect to observer when observer directly sees the object v_0 is :



6. A concavo-convex lens made of glass of refractive index 1.5 has surfaces of radii 20 cm and 60 cm . A concave mirror of radius of curvature 20 cm is placed co-axially to the lens. A glass slab of thickness 3 cm and refractive index 1.5 is placed closed to the mirror in the space between the mirror and lens as shown in figure. The distance of the nearest surface of the slab from lens is 248 cm . An object is placed 80 cm to the left of the lens. If final position of the image formed after refraction from lens, refraction from slab, reflection from mirror, refraction from the slab and lens is at the object O, find the distance x (in cm) of the mirror from nearer surface of slab.



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

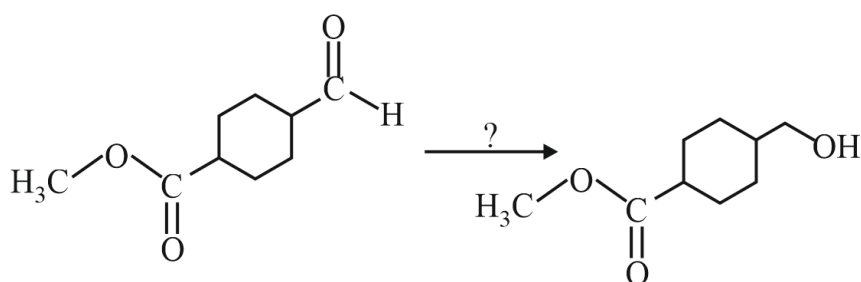
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Which of the following reagents can be used in the following reaction?



- (i) LiAlH_4
- (ii) NaBH_4
- (iii) $\text{H}_2/\text{Pd, BaSO}_4$
- (iv) $\text{NH}_2\text{-NH}_2/\text{OH}^-$
- (v) Zn-Hg/HCl

(A) i, ii, iii, iv

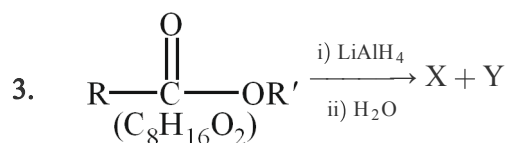
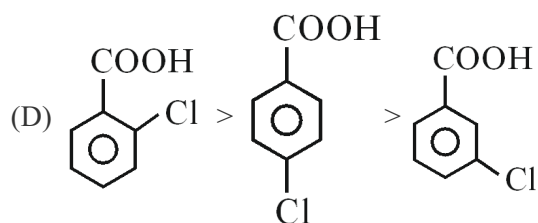
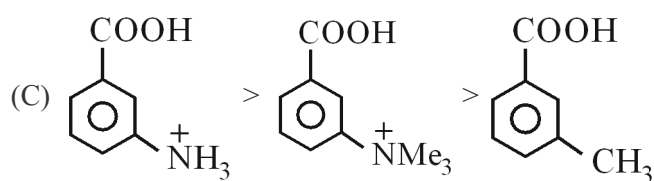
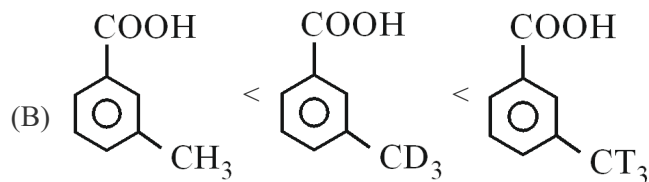
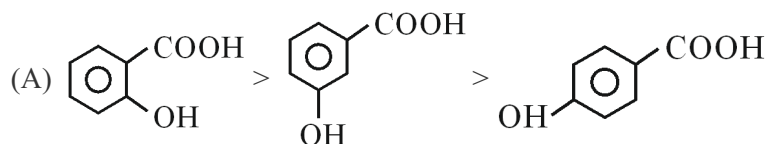
(B) ii, iii, v

(C) i, ii, iv

(D) ii

Space for Rough Work

2. Select the correct order of acid strength.



In Lucas test, turbidity is obtained instantly with 'Y', whereas 'X' fails to produce any turbidity. Which of the following statement will be true for – R & – R' alkyl groups?

(A) – R & – R' both are 3°

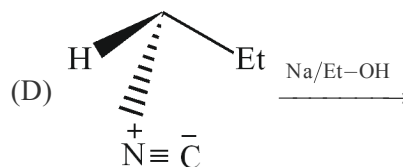
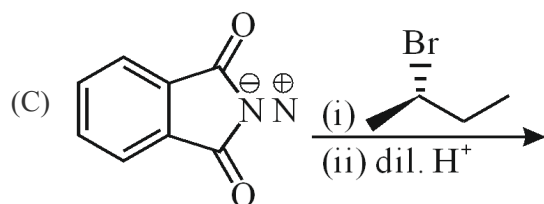
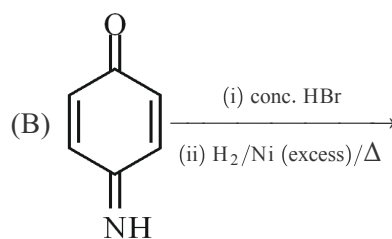
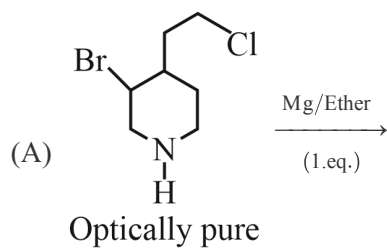
(B) – R must be 1° & – R' must be 3°

(C) – R must be 3° & – R' must be 1°

(D) – R & – R' both are 2°

Space for Rough Work

4. Identify which reaction produce single optically active amine molecule as product?



Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

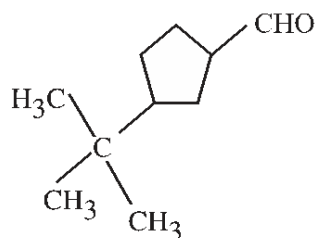
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

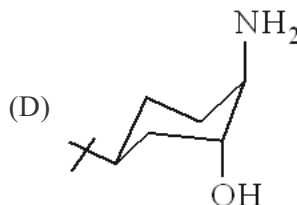
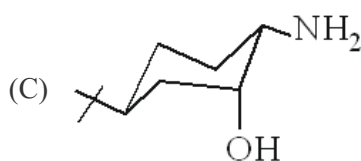
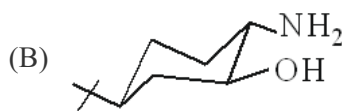
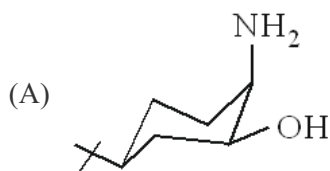
Negative Marks : -1 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. On reaction with NaNO_2/HCl , which of the following compounds will give the product 'X'.

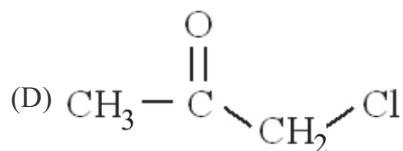
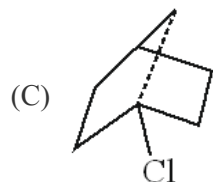
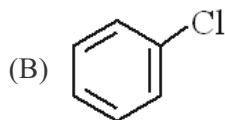
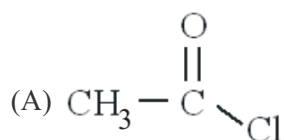


(X)

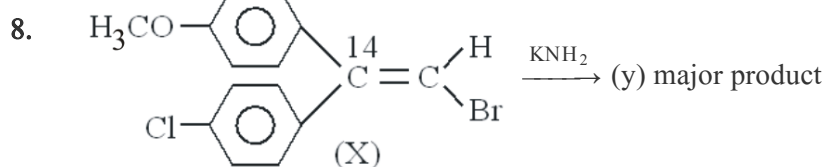
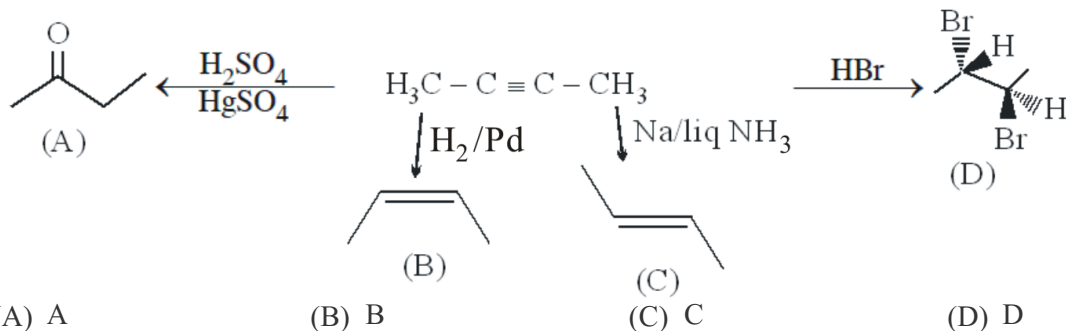


Space for Rough Work

6. Which of the following compounds can't be used for synthesis of Grignard reagent.

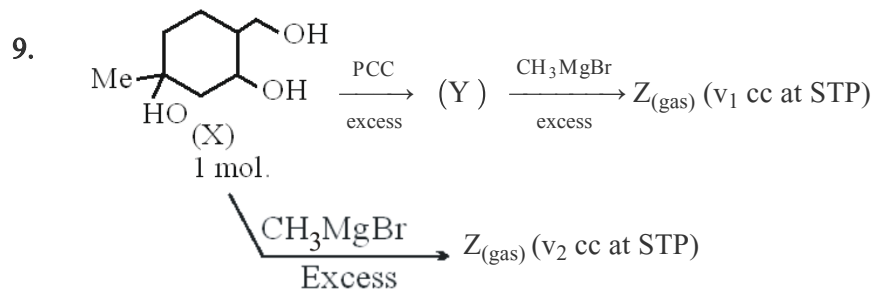


7. Choose the incorrect option(s):



- (A) The reaction involves rearrangement.
- (B) Final product is an alkyne.
- (C) Group anti to H will migrate.
- (D) In the final product C^{14} is connected directly to the chlorophenyl group.

Space for Rough Work



Choose the correct statement(s).

(A) $\frac{v_1}{v_2} = \frac{3}{1}$

(B) $\frac{v_1}{v_2} = \frac{1}{3}$

(C) Z is an alkane

(D) When Y reacts on reaction with NaHCO_3 , CO_2 gas is evolved.

10. Select the correct statement(s)

(A) Alkene is more reactive than alkyne towards electrophilic addition.

(B) Alkyne is more reactive than alkene towards hydrogenation.

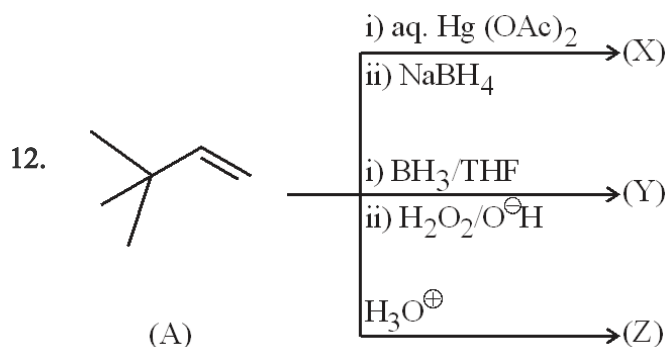
(C) Conjugated dienes undergo direct as well as conjugate addition.

(D) In conjugate addition, 1, 2, addition product is always less stable & form faster than 1, 4 addition product.

Space for Rough Work

11. Which of the following statement are incorrect?

- (A) All conforms of n-butane are achiral.
- (B) Gauche is the second most stable conformation of butane.
- (C) Percentage of the most stable conformer decreases when temperature is increased.
- (D) Dipole moment of a sample 1,2-dichloroethane is zero, even though same individual conformers are polar.



Correct statement X, Y and Z is/are

- (A) 'Z' upon oxidation with $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}^+$ give ketone
- (B) X and Y are chain isomers
- (C) Y when treated with only $\text{ZnCl}_2/\text{Conc. HCl}$ does not produce turbidity at room temperature immediately.
- (D) among X, Y and Z only 'X' contain a chiral carbon.

Space for Rough Work

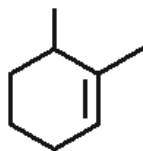
SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

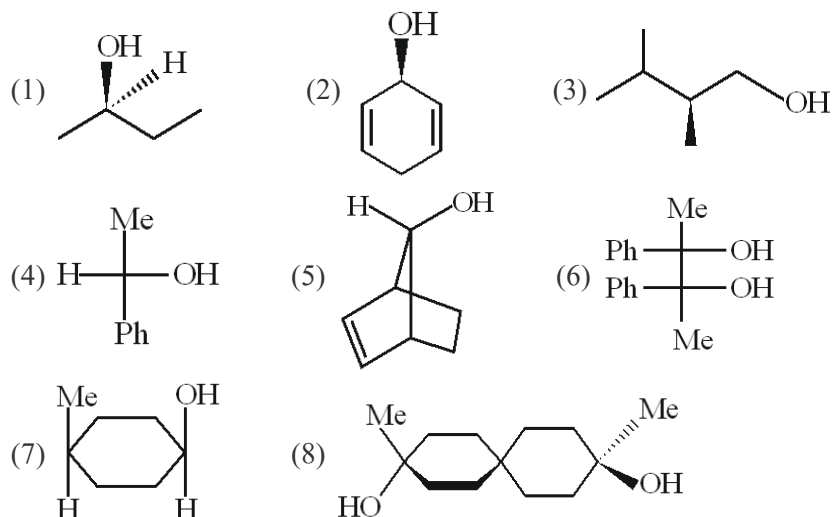
Zero Marks : 0 In all other cases.

1. An optically active compound A has the molecular formula C_6H_{10} . The compound gives a precipitate when treated with $Ag(NH_3)_2OH$. On catalytic hydrogenation, A yields $B(C_6H_{14})$ which is optically inactive. Identify the total number of α -H in product formed by treatment of A with O_3/H_2O_2 followed by treatment with $LiAlH_4$ and then heating with conc. H_2SO_4 .
2. Find the number of dichloro isomers of 2-methylbutane (including stereo isomers).
3. When the following compound is named according to IUPAC convention, what would be the sum of position of methyl groups.

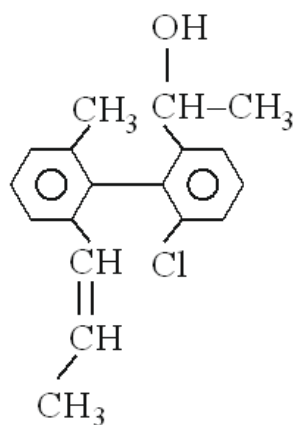


Space for Rough Work

4. How many of the following compound undergo racemization in an acidic medium?

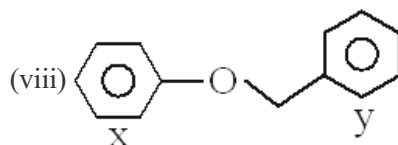
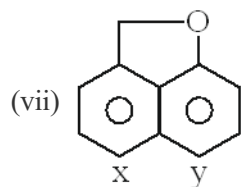
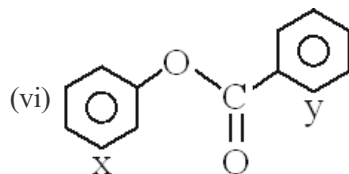
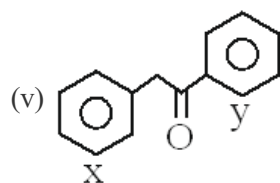
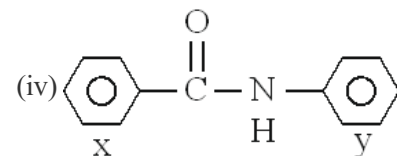
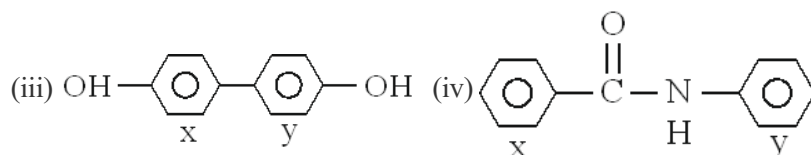
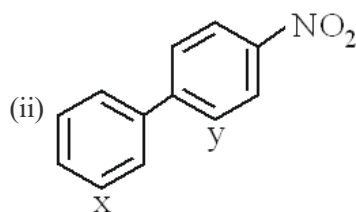
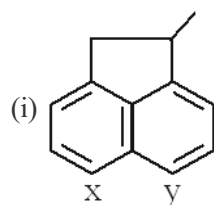


5. Total number of stereoisomers for the following compound is



Space for Rough Work

6. Identify the number of compounds in which ring labelled x is more active than ring labelled y for electrophilic aromatic substitution.



Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three vectors. If $|\vec{a}| = |\vec{b}| = 3$, $\vec{a} \cdot \vec{c} = 1$, $\vec{b} \cdot \vec{c} = 2$, $\vec{a} \cdot \vec{b} = 3$ and $[\vec{a} \ \vec{b} \ \vec{c}] = 3$. The projection of $8\vec{a} - 3\vec{b}$ on $\vec{c}$, is
 (A) $\frac{8}{\sqrt{13}}$ (B) $2\sqrt{\frac{24}{13}}$ (C) $\frac{2\sqrt{2}}{13}$ (D) $2\sqrt{\frac{12}{7}}$
2. Set A has 'm' elements and set B has 'n' elements if the total number of subsets of A is 112 more than the total number of subsets of B then the product of m and n is
 (A) 28 (B) 35 (C) 21 (D) 30
3. Let R be a relation on set $A = \{1, 2, 3, 4, 5\}$ defined as $R = \{(a, b) : |a^2 - b^2| < 8\}$. Then the relation R is
 (A) Reflexive, symmetric but not transitive (B) Transitive, symmetric but not Reflexive
 (C) Reflexive, transitive but not symmetric (D) Equivalence relation
4. The domain of $f(x) = \sqrt{3 - 2^x - 2^{1-x}} + \sqrt{1 - \sqrt{1 - x^2}}$, is
 (A) $[0, 1]$ (B) $[0, 2]$ (C) $[0, 3]$ (D) $[-\infty, 1]$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. A pyramid with vertex at point P, whose position vector $4\hat{i} + 2\hat{j} + 2\sqrt{3}\hat{k}$ has a regular hexagonal base ABCDEF. Position vectors of A and B are $\hat{i}$ and $\hat{i} + 2\hat{j}$ respectively, centre of the base has position vector $\hat{i} + \hat{j} + \sqrt{3}\hat{k}$. Then,
- (A) Position vector of foot of perpendicular from P to base surface is $\hat{i} - \hat{j} + (2\sqrt{3} - 3)\hat{k}$
- (B) Equation of line passing through vertex P and E; $\vec{r} = (\hat{i} + 2\sqrt{3}\hat{k}) + \lambda(3\hat{i} + 2\hat{j})$
- (C) Height of pyramid = $3\sqrt{3}$ unit
- (D) Foot of perpendicular from P to base surface lies on line AD.
6. If $A = \{x \in \mathbb{Z} : (x-1)^{(x+4)}(x^2+5x+6) = 1\}$ and $B = \{x \in \mathbb{Z} : (x^2-5x+5)^{(x^2+4x-60)} = 1\}$ where $\mathbb{Z}$ be the set of integers then which of the following is(are) **correct**?
- (A) The number of subsets of the set $A \times B$ are 2^{20}
- (B) The number of functions from set A to set B are 2^{10}
- (C) The number of injective functions from set A to set B are 120
- (D) The number of onto functions from set A to set B are 120

Space for Rough Work

7. Let $P_1: 2x + y + z = 3$, $P_2: 3x - y - 2z = 2$ be two planes. Then which of the following statement(s) is(are) correct?
- (A) Image of plane P_1 with respect to plane P_2 will be $5x + 10y + 13z = 15$
- (B) Plane P_1 rotated 90° about the common line of P_1 and P_2 then new equation of plane is $4x - 3y - 5z - 1 = 0$
- (C) Shortest distance between $L: \vec{r} = -\hat{i} + \mu(\hat{i} + \hat{j})$ and common line of P_1 and P_2 is $\frac{5}{\sqrt{114}}$
- (D) Shortest distance between $L: \vec{r} = -\hat{i} + \mu(\hat{i} + \hat{j})$ and common line of P_1 and P_2 is $\frac{7}{\sqrt{114}}$
8. Let $f(x) = \sin^{-1}\left(\frac{16-x^2}{16+x^2}\right) + \cos^{-1}\left(\frac{8x}{16+x^2}\right)$, then
- (A) Number of integers in the range of $f(x)$ are 4.
- (B) $\text{Sgn}(f(x)) = x^2 + 2x - 15$ has only two real solutions.
- (C) $f(x) = \frac{\pi}{4}||x| - 4|$ has only two real solutions.
- (D) $\left[\frac{4}{3}\cos(f_{\max}(x) + f_{\min}(x))\right] = -2$, $[\cdot]$ represent greatest integer function
9. If $\vec{b}$ and $\vec{c}$ be any two unit vectors such that $\vec{b} \cdot \vec{c} = 0$ and $\vec{a}$ be any vectors then
- (A) $(\vec{a} \cdot \vec{b})\vec{b} + (\vec{a} \cdot \vec{c})\vec{c} + [\vec{a} \ \vec{b} \ \vec{c}](\vec{b} \times \vec{c}) = \vec{b}$
- (B) $(\vec{a} \cdot \vec{b})\vec{b} + (\vec{a} \cdot \vec{c})\vec{c} + [\vec{a} \ \vec{b} \ \vec{c}](\vec{b} \times \vec{c}) = \vec{a}$
- (C) $(\vec{a} \cdot \vec{b} + 1)\vec{b} + (\vec{a} \cdot \vec{c})\vec{c} + [\vec{a} \ \vec{b} \ \vec{c}](\vec{b} \times \vec{c}) = \vec{a} + \vec{b}$
- (D) $(\vec{a} \cdot \vec{b})\vec{b} + (\vec{a} \cdot \vec{c} + 1)\vec{c} + [\vec{a} \ \vec{b} \ \vec{c}](\vec{b} \times \vec{c}) = \vec{a} + \vec{c}$

Space for Rough Work

10. Let $A = \{x: x \text{ is a 3 digit number}\}$
 $B = \{x: x = 9k + 2; k \in I\}$
 $C = \{x: x = 9k + 4; k \in I\}$
 If sum of elements in $A \cap (B \cup C)$ is 21900 λ then λ is(are)
- (A) A prime number (B) An even number
 (C) Composite number (D) An odd number
11. Let $\sum_{k=1}^{\infty} \sec^{-1} \left(\frac{\sqrt{k^2 + 4k + 3}}{1 + \sqrt{k^2 + 2k}} \right) = \alpha$, then (where $[\cdot]$ denotes the greatest integer function).
- (A) $[\sin 2\alpha + \cos 2\alpha] = -1$
 (B) $[\sqrt{6} \sin \alpha + 9 \sin 4\alpha] = -4$
 (C) $[\log_8 |\tan 2\alpha|] = -1$
 (D) $\sin^{-1} \left(\frac{2x}{1+x^2} \right) + \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right)$ at $x = \tan \left(\frac{3\pi}{4} - \alpha \right)$ is equal to π .
12. Suppose f, g, h be three real valued functions defined on R .
 Let $f(x) = 2x + |x|$, $g(x) = \frac{1}{3} (2x - |x|)$ and $h(x) = f(g(x))$.
- (A) Range of function $k(x) = \cos^{-1}(h(x)) + \cot^{-1}(h(x))$ is $\left[\frac{\pi}{4}, \frac{7\pi}{4} \right]$
 (B) $f(2f^{-1}(3)) + 3g^{-1}(2) = 60$
 (C) $f(g(h(h(x)))) = x \quad \forall x \in R$
 (D) Domain of function $I(x) = \sin^{-1} (f(x) - g(x))$ is $\left(-\infty, \frac{3}{8} \right]$

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

- Let $\vec{a}_r = x_r \hat{i} + y_r \hat{j} + z_r \hat{k}$ where $r = 1, 2, 3, 4$ such that $|\vec{a}_1 \times \vec{a}_2| = 3 |\vec{a}_3 \times \vec{a}_4| = 6$
and $(\vec{a}_1 \times \vec{a}_2) \cdot (\vec{a}_3 \times \vec{a}_4) = \frac{\pi}{3}$,
then $\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ 0 & 0 & 1 \end{vmatrix} \begin{vmatrix} x_3 & y_3 & z_3 \\ x_4 & y_4 & z_4 \\ 0 & 0 & 1 \end{vmatrix} - \begin{vmatrix} x_1 x_3 + y_1 y_3 + z_1 z_3 & x_1 x_4 + y_1 y_4 + z_1 z_4 & z_1 \\ x_2 x_3 + y_2 y_3 + z_2 z_3 & x_2 x_4 + y_2 y_4 + z_2 z_4 & z_2 \\ z_3 & z_4 & 0 \end{vmatrix}$ is
- If $f(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^5}{5}$ and $g(x) = \tan^{-1}x$, then number of solution(s) of the equation $\text{gof}(x) = f^{-1} \circ g^{-1}(x)$ is
- The distance of the point $(-2, 3, -4)$ from the line $\vec{r} = (3t - 2) \hat{i} + \left(2t - \frac{3}{2}\right) \hat{j} + \left(\frac{5}{3}t - \frac{4}{3}\right) \hat{k}$ measured parallel to the plane $4x + 12y - 3z + 1 = 0$, is

Space for Rough Work

4. Let $f(x) = |2x - |x + 1|| + |x - 1|$ and $g(x) = \frac{1 - x^2}{1 + x^2}$. If number of integers in the range of $f \circ g(x)$ and $g \circ f(x)$ are ℓ and m respectively, then $(\ell + m)$ is
5. If $S_n = \sum_{r=1}^n \tan^{-1} \left(\frac{32r}{16r^4 - 8r^2 + 17} \right)$ then $\sum_{n=1}^{10} \tan(S_n + \cot^{-1} 4)$ is equal to
6. If shortest distance between lines $\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1}$ and $\frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4}$ is d and the equation of the line of shortest distance is $\frac{x-3}{a} = \frac{y-c}{b} = \frac{z-3}{-1}$ then the value of $(a + b + c + d^2)$ is

Space for Rough Work

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

Space for Rough Work